NETFLIX CONTENT ANALYSIS

Natural Voiceover Script

Simple, clear spoken English • Beginner-friendly • Professional and conversational

Style reference: your uploaded EDA–Amazon Prime Video presentation. This script uses a similar teaching flow: introduce the question, explain the data, walk through the charts and modelling steps, then close with practical insights.

Accuracy note: the uploaded redpandacompress video shows a Netflix analysis notebook and machine-learning workflow. Some output values are too small to read reliably from the video, so this script does not invent totals, scores, or conclusions. Read any bracketed values from the actual notebook output before recording.

Suggested edited narration: around 6–8 minutes. You do not need to narrate every line of code; explain what each step does and why it matters.

1. Introduction | 0:00–0:35

Hello everyone. In this presentation, I'll walk you through my Netflix Content Analysis and Machine Learning project.

The aim is to explore the Netflix titles dataset, understand patterns in movies and TV shows, and then use machine learning to see whether information about a title can help predict its content type.

I'll explain the project step by step, from checking and cleaning the data to exploring charts, building a model, and discussing what the results mean.

> ***On-screen cue:*** *Show the project title and short project summary.*

2. Understanding the dataset | 0:35–1:10

First, I load the Netflix dataset into Python using pandas. The dataset contains information about titles available in the catalogue, such as the title name, content type, release year, rating, country, genre, duration, and description, depending on which fields are present.

I start by displaying the first few rows and checking the number of rows and columns. This gives me a quick overview of the information I have and helps me understand how the data is organized.

At this stage, I'm not making conclusions yet. I'm simply getting familiar with the dataset.

> ***On-screen cue:*** *Show the first rows, column names, and dataset dimensions. Read the actual row and column counts if clearly visible.*

3. Data quality and cleaning | 1:10–1:55

Before analysing the data, I check its quality. I look at the column names, data types, duplicate rows, and missing values.

Missing information is common in real datasets. For example, a title may not have a director or country listed. In this workflow, column names are standardized, exact duplicate rows are removed, and missing text values are marked as unknown where appropriate. The date-added field is converted to a date format when it is available.

These steps make the dataset more consistent. It is important to remember that cleaning does not mean making every value perfect; it means handling the issues carefully and transparently.

> ***On-screen cue:*** *Show the data information, missing-value summary, duplicate check, and cleaned dataset. Only mention counts you can read from the output.*

4. Exploring the catalogue with charts | 1:55–3:10

Now I use charts to understand the catalogue.

The first chart compares the number of movies and TV shows. It helps us see the balance between the two content types in this dataset.

Next, I look at release years to understand how the catalogue is distributed across time. The country chart shows which countries appear most often in the country information, and the genre chart highlights the categories that occur most frequently.

I also look at ratings, compare release years by content type, and explore the duration values. Each chart answers a different question and makes a large table easier to understand.

These charts describe the dataset we have. They should not automatically be treated as a complete picture of all Netflix content today, because the data reflects the source file and the period it represents.

> ***On-screen cue:*** *Show the charts one at a time. Briefly point out the visible pattern in each chart; avoid unsupported claims about causes or current Netflix catalogue size.*

5. Creating useful features | 3:10–3:45

After the initial exploration, I create a few additional features to make the information easier to use in analysis.

For example, the duration field may contain text such as minutes or seasons. I extract the numeric part into a separate duration value. I also count how many genres and countries are listed for a title when that information is available.

This is called feature engineering. In simple terms, it means turning existing information into a format that is more useful for analysis or machine learning.

One caution is that a duration number can mean minutes for a movie or seasons for a TV show, so it needs to be interpreted carefully.

> ***On-screen cue:*** *Show the feature-engineering code and the new columns. Do not imply that a duration value means the same thing for both content types.*

6. The machine-learning question | 3:45–4:20

The machine-learning part asks whether selected title details can help predict if a title is a movie or a TV show.

The target is the content type. The input features used in the workflow include available fields such as release year, rating, duration value, genre count, and country count.

Because the answer is one of two categories, this is a classification problem. I split the data into training and testing sets. The training data is used to teach the model, while the testing data is held back so I can evaluate its predictions on examples it has not trained on.

> ***On-screen cue:*** *Show the target, selected features, and train-test split. Read the actual split sizes only if visible.*

7. Comparing models | 4:20–5:20

I compare two classification approaches: Logistic Regression and Random Forest.

Logistic Regression is a commonly used baseline for classification. Random Forest combines many decision trees, allowing it to learn more complex patterns in the data.

Before training, the workflow prepares the features. Missing numeric values are filled using the median, numeric values are scaled for Logistic Regression, and text categories such as ratings are converted into numeric indicators using one-hot encoding.

I then compare the models using accuracy, precision, recall, and F1-score. Accuracy measures the overall share of correct predictions. Precision tells us how reliable positive predictions are, recall tells us how many examples of a class the model identifies, and F1-score balances precision and recall.

The model comparison table and confusion matrix should guide the explanation. Here, I'll report the values produced by my run: [read the actual best model and metrics from the output]. I won't assume one model is better unless the displayed results support that conclusion.

> ***On-screen cue:*** *Show the model comparison table and confusion matrix. Replace the bracketed sentence with actual output values from the notebook.*

8. Interpreting the results responsibly | 5:20–6:00

The confusion matrix gives a more detailed view of the predictions. It compares the actual class with the class predicted by the model, so we can see which movies or TV shows were classified correctly and where the model made mistakes.

These results show how well the model performs on this particular test split. They do not prove that it will perform equally well on every Netflix title or on a different dataset.

The selected features are limited, and some fields may be missing or inconsistent. Also, the project predicts content type; it is not a recommendation system and does not predict whether a title will be popular.

> ***On-screen cue:*** *Point to the confusion matrix and explain the axis labels. Keep the distinction between classification and recommendation clear.*

9. Practical meaning and future improvements | 6:00–6:45

This analysis can be useful as a learning example for catalogue exploration, metadata checks, and understanding how machine learning works with structured data.

There are several ways to extend the project. We could build a recommendation feature based on title descriptions, create an interactive dashboard, tune the model parameters, or use tools that explain which features influenced a prediction.

We could also add stronger data-quality checks and test the model on more recent or more representative data. These are possible next steps, rather than features I'm claiming are already completed in this version.

> ***On-screen cue:*** *Show the project conclusion or future-work section. Label suggested extensions as future work.*

10. Conclusion | 6:45–7:10

To conclude, this project follows a complete data-analysis workflow: inspect and clean the data, explore it with charts, create useful features, train classification models, and evaluate their predictions.

The main lesson is that good analysis is not just about building a model. It is also about understanding the data, choosing suitable evaluation measures, and being honest about limitations.

Thank you for watching my Netflix Content Analysis and Machine Learning project.

> ***On-screen cue:*** *Finish on the conclusion screen. If your GitHub repository is ready, show its name or link briefly.*

Recording tips

- Speak slowly and pause briefly between sections.
- Zoom in so code outputs and chart labels are readable.
- Do not read every line of code; explain the purpose of each step.
- Run the notebook before recording so the charts and metric outputs are visible.
- Replace bracketed prompts with the exact values from your own run. If you cannot verify a value, leave it out.